

MITHIL K GOWDA

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Professional Summary

Cybersecurity graduate specializing in Application Security, Secure Software Development, Security Architecture, API Security, and Web Application Security. Project Intern at ISRO-LEOS with experience designing and developing secure backend applications using Python, Flask, REST APIs, and MySQL. First author of internationally recognized cybersecurity research focused on secure architectures, adaptive cyber defense, cryptographic security, and AI driven threat detection. Passionate about designing resilient applications and integrating security throughout the software development lifecycle.

Education

M.S. Ramaiah University of Applied Sciences, Bengaluru

2022–2026

B.Tech. in Information Science and Engineering

CGPA: **8.80 / 10**

Technical Skills

Application Security: Secure Software Development, Secure SDLC, Security Architecture, Authentication, Authorization, API Security, Web Application Security, Threat Modeling, Secure Coding, OWASP Top 10

Application Security Testing: Burp Suite, OWASP ZAP, Nmap, Nessus, Metasploit, Vulnerability Assessment

Programming: Python, Java, JavaScript, SQL, HTML, CSS, Bash

Frameworks: Flask, Node.js, REST APIs

Cryptography: AES-256, SHA-256, Zero-Knowledge Proofs (ZKP), Secure Transaction Verification

Databases: MySQL, MongoDB

Cloud: Google Cloud Platform (GCP)

Operating Systems: Linux, Windows

Professional Experience

Indian Space Research Organisation (ISRO) – Laboratory for Electro Optics Systems (LEOS), Bengaluru
2026

Project Intern – Secure Software Development & Data Analytics

- Designed and developed secure backend applications using Python, Flask, REST APIs, and MySQL following secure software engineering principles.
- Implemented authentication, authorization, secure database interaction, and protected application workflows to strengthen application security and controlled access.
- Built secure data processing and visualization pipelines supporting operational analytics, structured data management, and engineering decision-making.
- Designed maintainable backend services emphasizing secure architecture, modular development, secure API design, operational reliability, and protected application communication.

Application Security Projects & Research

A Multi-Agent AI Cyber Defense Framework Using HoneyBee Method

- First Author and Corresponding Author of an AI-driven cybersecurity framework accepted for presentation at ICIICE 2026 (Dubai, UAE) and accepted as a book chapter in the edited volume *Metaheuristic Optimization for Social Good*.
- Designed a secure multi-agent cyber defense architecture integrating adaptive security, intelligent threat detection, resilient application security, and autonomous defensive mechanisms.
- Engineered the Detect–Mislead–Neutralize–Learn framework integrating XGBoost, CNN-LSTM, Deep Q-Network (DQN), adaptive honeypots, distributed threat intelligence, and secure system architecture.
- Evaluated the framework using CIC-IDS2017 datasets and simulated enterprise attack scenarios, achieving 98.24% detection accuracy, 98.51% precision, 98.10% recall, 0.992 ROC-AUC, and 1.85% false positive rate.

AI-Governed Carbon Credit Exchange with Digital Carbon Passport

- Abstract accepted at the 4th International Conference on Advancing Sustainable Futures (ICASF 2027), Abu Dhabi University, UAE.
- Designed a secure distributed application integrating AES-256 encryption, SHA-256 hashing, authentication mechanisms, fraud detection, trust scoring, and secure transaction validation for digital asset protection.
- Developed the Secure Trade Authorization and Verification Pipeline (STAVP) utilizing AES-256 encryption, SHA-256 hashing, authentication mechanisms, and secure transaction verification.

C³T-STAVP Cryptographic Framework (Ongoing Research)

- Designing a hybrid cybersecurity framework integrating Character Chaffing & Transposition Technology (C³T), Secure Transaction Authorization & Verification Protocol (STAVP), and adaptive trust validation.
- Researching Zero-Knowledge Proofs (ZKP), post-quantum cryptography, semantic obfuscation, privacy-preserving architectures, secure transaction validation, and resilient application security mechanisms.

Professional Certifications

- IBM IT Fundamentals for Cybersecurity Specialization – IBM
- Ethical Hacker – Cisco Networking Academy
- Systems and Usable Security (Elite) – NPTEL
- IBM Front-End Developer Specialization – IBM
- Introduction to Cyber Security – IGNOU (SWAYAM)
- Google Cloud Computing – NPTEL
- Introduction to Internet of Things (Silver Medal) – NPTEL
- Business Intelligence and Analytics – NPTEL

Achievements

- Selected as Project Intern at the Indian Space Research Organisation (ISRO – LEOS).
- First Author of internationally recognized cybersecurity research focused on AI-driven cyber defense, autonomous threat detection, and intelligent security architectures.
- Awarded the Reliance Foundation Undergraduate Scholarship (2026).
- Recipient of NPTEL Silver Medal in Introduction to Internet of Things.